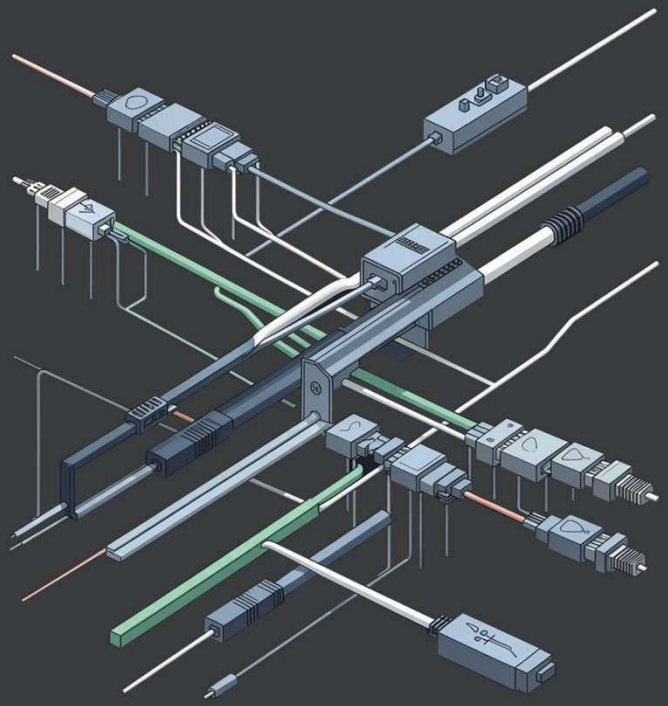


What is Parity?

Error Detection

Technique for catching errors in data transmission

Extra Bit

Adds one bit to detect single-bit errors

Even & Odd Parity

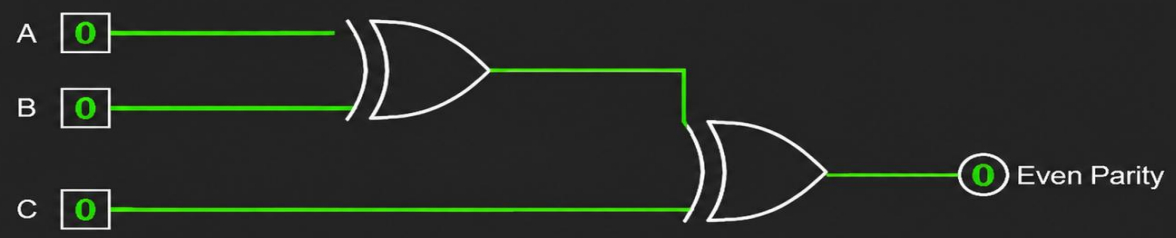
Even Parity

Total 1s including parity bit = even number

Odd Parity

Total 1s including parity bit = odd number

EVEN PARITY CIRCUIT



A	B	C	EvenParity
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Made with GAMMA

ODD PARITY CIRCUIT



A	B	C	OddParity
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Made with GAMMA

XOR Gate Truth Table

Output is 1 when inputs differ, 0 when inputs are the same.

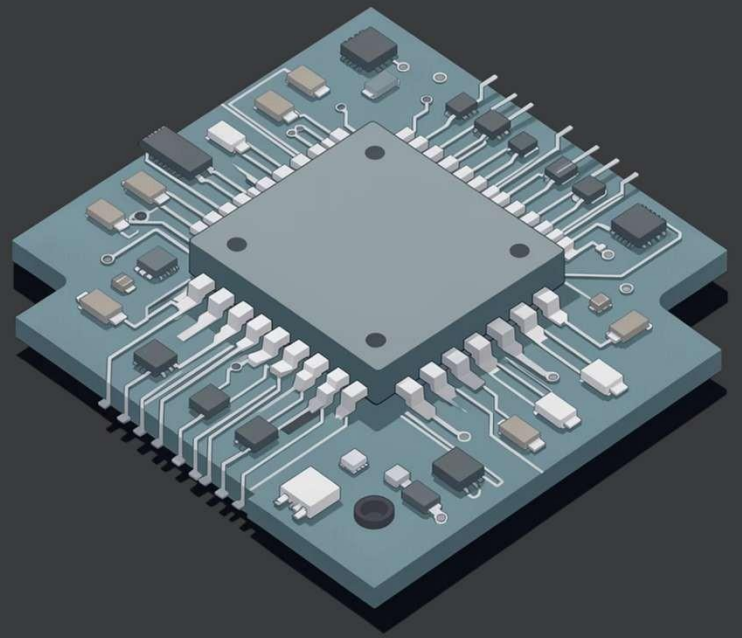
XOR is the core logic behind all parity computation.

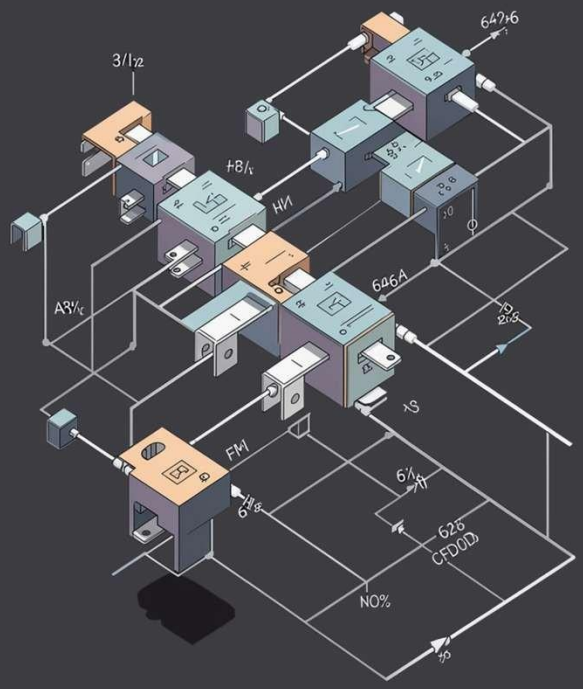
A	B	Out
0	0	0
0	1	1
1	0	1
1	1	0

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Parity Bits & XOR Gates in Digital Systems

Error detection fundamentals in digital logic design





Why XOR for Digital Logic?

Parity Logic

Implements checking natively

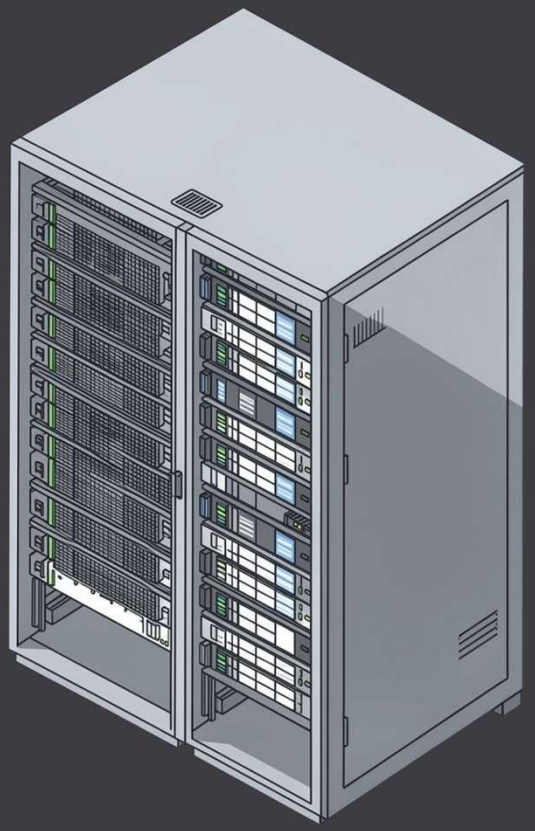
Hardware Efficient

Fast, minimal gate count

Versatile

Error detection & cryptography

Made with GAMMA



Real-World Applications

Memory

RAM error detection

Communication

RS-232, USB protocols

Storage

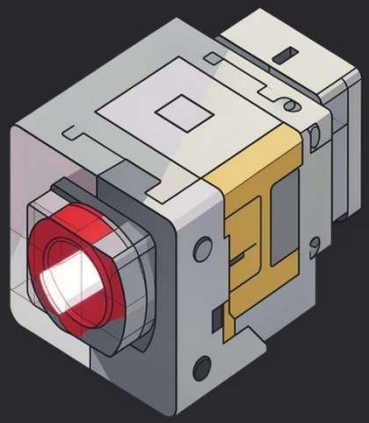
Hard drives, SSDs

Networks

Ethernet frames

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Parity Checker Circuit



Verifies parity correctness

Re-computes parity on received data

Error signal output

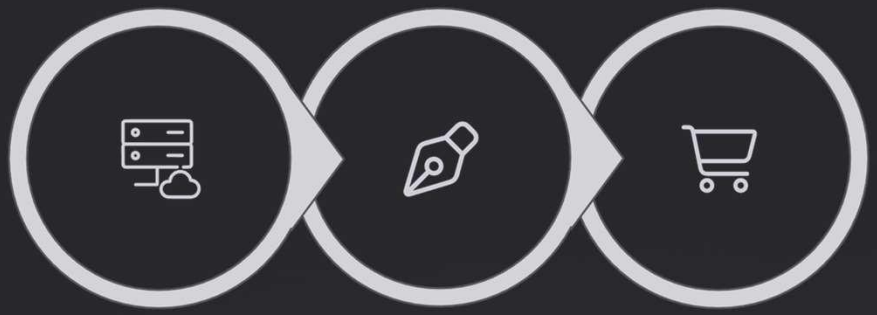
Flags mismatch between expected and received parity

Detects transmission errors

Catches single-bit corruption in transit

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What is a Parity Generator?



Input Bits XOR Array Parity Bit

- **Calculates parity bit**
  Circuit output appended to data
- **Built from XOR gates**
  Chained together internally
- **Real-time operation**
  Computes on the fly during transmission

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Odd Parity Generator

1

Goal

Make total count of 1s odd

2

Count is Even

Parity bit =

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Even Parity Generator

1

Goal

Make total count of 1s even

2

Count is Odd

Parity bit =

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